

Capacitor C2 is used for further filtering input voltage at point D (pin 3 of op-amp). ZD1 is used as a protective device for suppressing voltage beyond 15V during unusual conditions. LED2 is connected in parallel with the relay, and is used to indicate device off or circuit open status. LED1 is connected at output pin 6 of IC1. It is used to indicate output sense. It is also used for setting load current.

When load current is more than the set value, LED1 glows. Two neon bulbs (B1 and B2) are used to indicate 230V AC input and AC output. AC output connector CON3 is used to connect the load.

Test point TP1 provides DC voltage corresponding to load current. Two resistors (R3 and R4) are chosen

to facilitate current measurement from a low value, that is, 300mA, to a high range of 5A. Developed voltages at TP1 and TP2 corresponding to current via L1 winding are given in Table II.

TP1 and TP2 are connected to points A and B of SPDT switch S2. Its middle pin D is connected to pin 3 of IC1. S2 is a range-selection switch. When it is at position A, low range is connected to pin 3 of IC1. When it is at position B, high range is connected to pin 3 of IC1. Table I shows the voltage developed during load current.

Voltage at pin 3 of IC1 acts as comparing voltage. Potmeter VR1 provides the reference voltage for comparison, which varies from 2V to 11V. VR1 is used for load current adjustment. It is

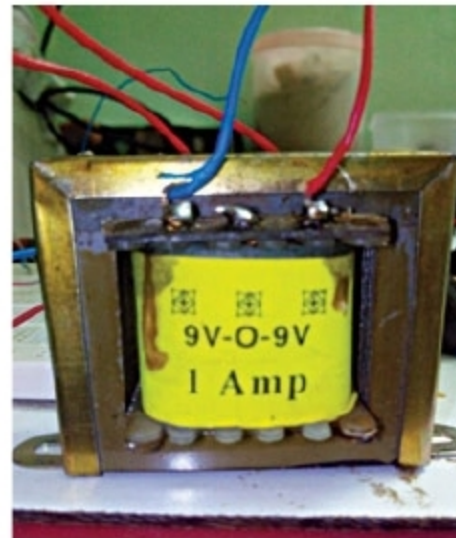


Fig. 3: Before winding on X1



Fig. 4: After winding (L1 side) on bobbin of X1

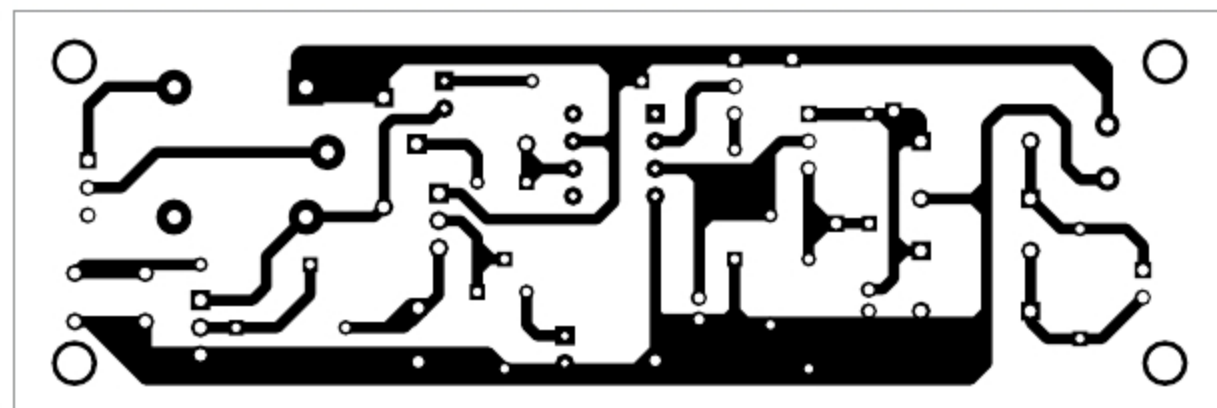


Fig. 5: Actual-size PCB layout of adjustable AC circuit breaker

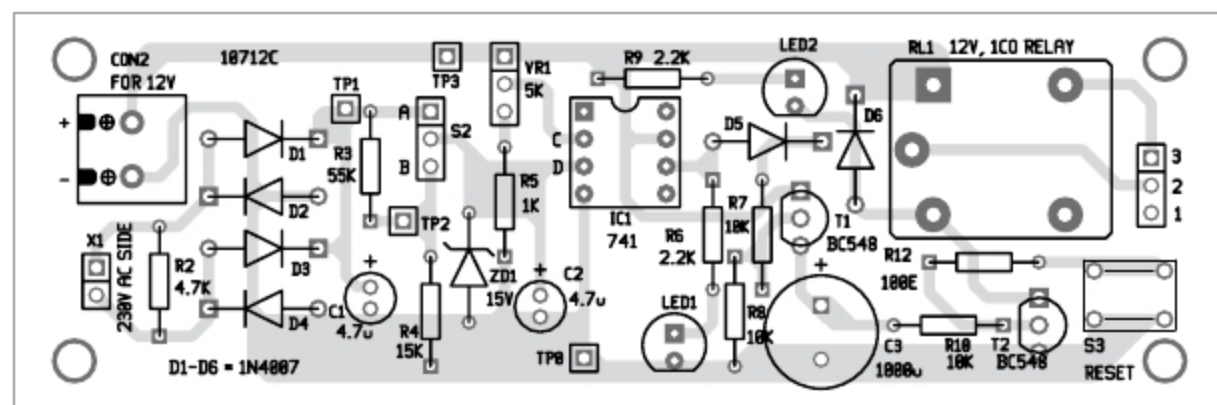


Fig. 6: Components layout for the PCB

TABLE I
VALUES OF VOLTAGES*
CORRESPONDING TO LOAD CURRENT
(IL) FLOWING THROUGH WINDING L1

IL through L1 winding (A)	Voltage at point A (V)	IL through L1 winding (A)	Voltage at point B (V)
0.14	0.99	1	2.44
0.5	5.05	2	5.33
1.0	9.57	3.5	8.61
1.3	10.73	4.5	9.76
1.5	11.29	5.5	10.46
1.8	11.9	6	10.8

*May vary depending on number of turns and value of R2

TABLE II
TEST POINTS

Test point	Voltage
TP0	Gnd
TP1	2V to 11V depending on load current
TP2	1/3 of voltage at TP1
TP3	12V

varied to set the reference voltage at point C (pin 2) of IC1. When voltage at pin 3 of IC1 exceeds set reference voltage at pin 2 due to current through L1, output at pin 6 of IC1 goes high and triggers relay RL1. Due to this, 230V AC output is switched off, and load current flowing through L1 is cut off.

A delay circuit is provided between IC1 and relay circuit. T1, C3 and R10 provide the delay. When output of IC1 goes high at pin 6, T1 conducts and charges C3 to maximum voltage (near 12V). D5 is used to pass the current in one direction, towards the base of T1.

C3 is discharged only through T2 via R10. T2 is used to drive RL1. R10 and C3 provide maximum delay of 10 seconds approximately. S3 is reset and connected to base of T2 through R12 to ground. When S3 is closed, it switches off T2, RL1 gets de-energised and its pole contact returns to its NC position.

Construction and testing

An actual-size PCB layout of the adjustable AC circuit breaker is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit on the PCB, connect external 12V across CON2. Note that, CON1 and CON3, S1, F1, and B1 and B2 are not